Detecting and Mitigating Semantic Drift in Multi-Turn Instruction-Based Image Editing

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***Abstract*—Instruction-based image editors such as InstructPix2Pix let users apply a sequence of natural-language edits to an image, yet no standard metric captures what happens to the parts of the image nobody asked to change. This paper introduces the Drift Score, a model-agnostic metric that combines Segment Anything (SAM) region segmentation with CLIP embedding similarity to quantify unintended change after an edit. The metric is applied across 120 four-step edit chains built from 60 COCO images, split evenly between localized object-level edits and broad scene-level edits. Two low-cost mitigation strategies, masked conditioning and region-locking, are implemented and evaluated against an unmitigated baseline using paired significance testing. Both mitigations produce large, highly significant reductions in measured drift (region-locking: 89.1%, $p < 0.0001$; masked conditioning: 68.6%, $p < 0.0001$). Two pre-registered hypotheses did not hold as predicted: region-locking significantly outperforms masked conditioning rather than the reverse, and drift does not increase across the chain, with the only significant step-position effect running opposite to the expected compounding pattern. Both reversals are traced to specific mechanisms rather than left unexplained. The Drift Score, dataset, and full pipeline are released as a reproducible toolkit built entirely from pretrained models.**

***Index Terms*—Image Editing, Diffusion Models, Semantic Drift, CLIP, Segment Anything, Evaluation Metrics, Instruction-Based Editing**

I. Introduction

Iterative refinement is a natural way to use an instruction-based image editor. A user might ask the model to make the sky look like sunset, then add a bird, then remove a fence in the background, refining an image through several small edits rather than one large prompt. This pattern introduces a failure mode that single-step evaluation cannot see: a model can follow every individual instruction correctly while still drifting further and further from what the user actually wanted, simply because nothing in its training or inference constrains it to leave the rest of the image alone. Existing evaluation protocols for these models, including CLIP-based instruction-following scores and human preference studies, measure whether the requested change happened. None of them measure what else changed as a side effect. This paper calls that side effect *semantic drift*.

Most existing instruction-based editors, InstructPix2Pix chief among them [1], are trained on synthetically generated (image, instruction, edited-image) triplets. That training recipe supplies no explicit signal for "leave everything else untouched" beyond whatever the data-generation process happened to preserve. Drift is therefore an expected consequence of how these models are built, not merely an empirical curiosity worth measuring after the fact.

Three questions follow directly from this observation. Can unintended drift be measured automatically, without manual inspection of every edited image (RQ1)? Does drift accumulate as a chain of edits gets longer, so that later instructions cause more collateral damage than earlier ones (RQ2)? And can cheap, pretrained-model-only interventions reduce it (RQ3)? This paper answers all three.

The contribution is threefold. First, the Drift Score, a reusable metric combining Segment Anything (SAM) region segmentation with CLIP embedding similarity to quantify unintended regional change, at both the single-edit and full-chain level. Second, an empirical study of how drift behaves across 120 real four-step edit chains, split between localized object-level edits and broad scene-level edits. Third, a statistically grounded comparison of two mitigation strategies, masked conditioning and region-locking, against an unmitigated baseline.

The work was scoped for a single contributor on a 30-day timeline using only pretrained models and free-tier compute: a laptop for development and analysis, and a free Google Colab GPU for the heavier inference stages. No model was trained or fine-tuned. The contribution here is the measurement methodology and the empirical findings it produces, not a new editing model.

A. Foundations

InstructPix2Pix [1] is the editor evaluated throughout this study. It fine-tunes a latent diffusion model [2] on a large corpus of synthetically generated editing triplets, produced by combining a language model with Prompt-to-Prompt-style attention injection [3]. Because supervision comes entirely from this generated data, the model inherits no explicit signal for preserving untouched content beyond what that generation process happened to keep stable, which is the mechanistic root of the drift problem this paper measures. CLIP [4] supplies the embedding space the Drift Score measures change in, and the text-image similarity used to identify which region an instruction targets. Segment Anything [5] supplies the region decomposition without which drift could only be measured at whole-image granularity, obscuring exactly the localized side effects this paper is concerned with.

B. Consistency-Preserving Editing

Prompt-to-Prompt [3], Plug-and-Play [6], and MasaCtrl [7] are generation-time techniques for keeping an edited image consistent with its source, by manipulating cross-attention maps, injecting spatial features, or converting self-attention into a mutual query against the source image's own generation trajectory. Null-text Inversion [9] extends Prompt-to-Prompt to real photographs by inverting the input image into the model's diffusion trajectory and optimizing only the unconditional embedding used for classifier-free guidance, avoiding the fine-tuning that competing inversion methods need. Imagic [10] achieves complex non-rigid edits on a single real image by optimizing a text embedding and then fine-tuning the diffusion model around it, a per-image cost that is incompatible with running many short edit chains under a fixed compute budget, one reason this study uses a forward-pass editor instead. All four methods aim to *produce* a more consistent single edit. None of them measure how much unintended change remains afterward, and none address a chain of sequential edits. MasaCtrl's own motivating discussion states the semantic-drift problem in almost the same terms used here, but stops at fixing one edit rather than measuring or chaining it.

C. Instruction-Following Accuracy

A related but distinct line of work targets not the model touching the wrong region, but the model misreading the instruction itself. Emu Edit [11] trains a single model across sixteen editing and vision tasks with learned task embeddings, improving both instruction adherence and general output quality relative to InstructPix2Pix. HIVE [12] instead collects human rankings of candidate edits and fine-tunes the base model on a learned reward signal, in a pipeline modeled on reinforcement learning from human feedback for language models. Both report large gains in whether the requested edit happened as asked. Neither reports what else changed as a side effect, which is the axis this paper measures; instruction-following accuracy and semantic drift are related but separate properties of an edit, and improving one does not by construction improve the other.

D. Multi-Turn Editing Data

MagicBrush [8] already contains manually annotated multi-turn editing sessions, which is independent validation that a four- or five-step chain is a realistic usage pattern rather than an artificial stress test. Its evaluation protocol, however, only checks whether each turn's target region changed the way it was supposed to. It never measures how much everything else changed in the process, which is exactly the gap this paper addresses.

TABLE I
CAPABILITY COMPARISON OF RELATED APPROACHES

Approach	Unintended Change	Chain Aware	Mitigation Compare	Model Agnostic
Prompt-to-Prompt [3]	×	×	×	×
Null-text Inv. [9]	×	×	×	×
Plug-and-Play [6]	×	×	×	×
MasaCtrl [7]	×	×	×	×
Emu Edit [11] / HIVE [12]	×	×	×	×
MagicBrush [8]	Partial	✓	×	×
Proposed (Drift Score)	✓	✓	✓	✓

No prior approach measures unintended regional change across a chain of edits or compares mitigation strategies on that specific axis (Table I); these dimensions are author-defined for this comparison rather than drawn from a standardized benchmark. The contribution of this paper is that combination: a measurement instrument built on top of the segmentation and embedding tools the field already uses, applied to the chain-based usage pattern the field already knows is realistic.

II. METHODOLOGY

The pipeline realizes a Segment → Identify → Edit → Score cycle at every step of a chain (Fig. 1), repeated once per instruction.

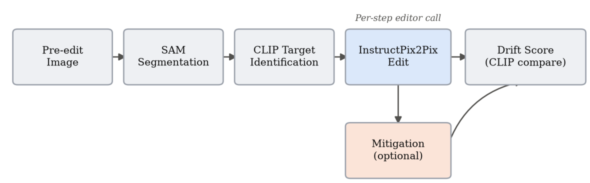

Fig. 1. Per-step pipeline. The mitigation branch is absent for the unmitigated baseline and active for the two mitigation strategies.

A. The Drift Score

Given a pre-edit image, a post-edit image, and the instruction that produced it, the pre-edit image is first segmented into regions with SAM, and each region is cropped from both the pre- and post-edit image at the *same* coordinates. Segmenting the post-edit image independently would yield unrelated region identifiers with no correspondence between the two images, so the same box set is deliberately reused. The target region, the one box the instruction most plausibly refers to, is identified by embedding every region crop and the instruction text with CLIP and taking the highest cosine similarity. For every non-target region r, the per-region drift is

$$d_r = 1 - \cos\left(\mathrm{CLIP}(x_r^{\mathrm{pre}}),\ \mathrm{CLIP}(x_r^{\mathrm{post}})\right) \tag{1}$$

and the per-step Drift Score is the mean of d_r over all non-target regions. A chain's cumulative Drift Score is the sum of its per-step scores:

$$D_{\text{chain}} = \sum_{t=1}^{T} \frac{1}{|R_t|} \sum_{r \in R_t} d_{r,t} \tag{2}$$

where R_t is the set of non-target regions at step t and T is the chain length. Higher values indicate more unintended change accumulated across the chain.

B. Dataset

Sixty images were sampled from COCO val2017 (seed 42, reproducible), filtered to those with two to eight annotated objects, enough to guarantee both a plausible target region and several non-target regions without an overly cluttered scene. Every image was inspected individually and given two handwritten, per-image instruction chains rather than instructions drawn from a fixed template pool:

- **Chain A (object-level)**: four localized, single-object edits, for example changing a horse's color and removing an obstacle in front of it.
- **Chain B (global)**: four broad, scene-level edits, for example making a scene rainy and adding sunset lighting.

This produced 120 chains in total: 60 object-level and 60 global, each four instructions long.

C. Pipeline and Models

TABLE II
MODELS USED, ALL PRETRAINED AND PUBLICLY AVAILABLE

Component	Model	Role
Editor	InstructPix2Pix	Executes each edit, 512×512
Segmentation	SAM (ViT-B)	Region decomposition
Embedding	CLIP (ViT-B/32)	Change measurement, targeting

No model was trained or fine-tuned. Heavy inference, editing and segmentation, ran on a free-tier Colab GPU; drift-score aggregation and statistical analysis ran locally on CPU.

D. Mitigation Strategies

Two strategies were implemented, both identifying the target region through the identical SAM-plus-CLIP procedure the Drift Score itself uses, so that measurement and mitigation agree on what counts as the target. They were chosen to sit at two different points on the same intervention spectrum, one correcting the model's output after the fact and one constraining what the model is given to work with in the first place, so that comparing them says something about *when* intervening is more effective, not just *whether* intervening helps at all.

Region-locking runs the edit on the full image as usual, then reverts every pixel outside the target region's box back to the pre-edit image. This is a correction applied after generation.

Masked conditioning crops the image down to just the target region, with a 32-pixel padding margin included for surrounding context, before generation. The margin exists because a crop tight to the target box alone tends to strip away the visual context InstructPix2Pix needs to render a coherent edit, at the cost, examined in Section IV-B, of extending what counts as "touched" slightly beyond the target box itself. The edit is applied only to that crop, and the result is pasted back into the full frame. The model never sees the rest of the image, which makes this a constraint applied at generation time rather than afterward.

A third strategy considered during scoping, attention-restricted editing informed by Prompt-to-Prompt-style cross-attention control, was treated as optional and was not implemented, since RQ3 already has a clear, statistically supported answer without it.

E. Statistical Analysis

Paired t-tests and Wilcoxon signed-rank tests compare cumulative Drift Scores between conditions, paired by image identity and chain type. The Wilcoxon test is treated as the primary result given the small, likely skewed sample; the t-test is reported alongside it for completeness. Of 480 baseline edit steps, 13 (2.7%) could not be scored, because SAM occasionally finds too few regions in a heavily stylized pre-edit image for anything to remain once the target region is excluded. These were dropped from pairwise comparisons rather than imputed.

III. EXPERIMENT AND EVALUATION

A. Experimental Setup

All 120 chains were run three times: once unmitigated (baseline), once with region-locking active at every step, and once with masked conditioning active at every step. Each run produced five images per chain, the original plus one output per instruction, all committed alongside their Drift Scores for reproducibility. The baseline chain outputs also served as the pre-edit input for the corresponding mitigated chains at each step, so all three conditions edit from the same starting point at every stage.

B. RQ1: Validating the Drift Score

Before trusting the score at scale, chains spanning the full score range were inspected by eye. The lowest-scoring baseline chain (0.082, a surfer riding a wave, Fig. 2) shows only the requested outfit and surfboard color changes; the wave, spray, and sky are the same scene, pixel for pixel, outside those two objects, exactly the behavior a near-zero score should describe.

Fig. 2. Lowest-scoring baseline chain. Only the outfit and surfboard color change; the wave, spray, and sky are unaffected.

The highest-scoring baseline chain (0.565, a boy holding a baseball glove) is a genuine catastrophic failure, and its per-step trajectory (Table III) is informative in its own right: drift climbs sharply from 0.074 at step 1 to 0.213 at step 2, then to 0.278 at step 3, at which point the image has collapsed to near-total black after the second edit destroyed the original composition. Step 4 could not be scored at all, since too little of the image remained for SAM to find a usable set of regions once the target was excluded, so the reported cumulative 0.565 is actually a sum of only three measurable steps, not four; the true drift is plausibly higher still. The fourth instruction was still applied on top of the wreckage (Fig. 4, second panel). The score correctly identifies this chain as the single worst in the dataset, which is the kind of agreement between the metric and visible content that a purely numerical validation cannot substitute for.

TABLE III
PER-STEP DRIFT, FLAGSHIP CHAIN (BASEBALL GLOVE)

Condition	Step 1	Step 2	Step 3	Step 4
Baseline	0.074	0.213	0.278	n/a[a]
Region-Locking	0.006	0.004	0.000	0.066
Masked Conditioning	0.018	0.002	0.029	0.132

[a]Unscoreable; excluded from the reported cumulative total.

Both mitigations keep steps 1 through 3 close to zero, confirming they suppress drift throughout the chain, not only on average. Both also spike at step 4, the "add a cap" instruction, for the reason traced in Section IV-C: an oversized target box leaves neither mitigation with a meaningful constraint on that specific step.

C. RQ3: Mitigation Effectiveness

Both mitigations reduce mean cumulative drift by a wide margin relative to the unmitigated baseline (Table IV, Fig. 3), and the reduction is significant well beyond conventional thresholds in every breakdown tested.

TABLE IV
CUMULATIVE DRIFT SCORE BY CONDITION ($n = 120$)

Condition	Mean	Reduction	Wilcoxon p
Baseline	0.293	–	–
Masked Conditioning	0.092	68.6%	< 0.0001
Region-Locking	0.032	89.1%	< 0.0001

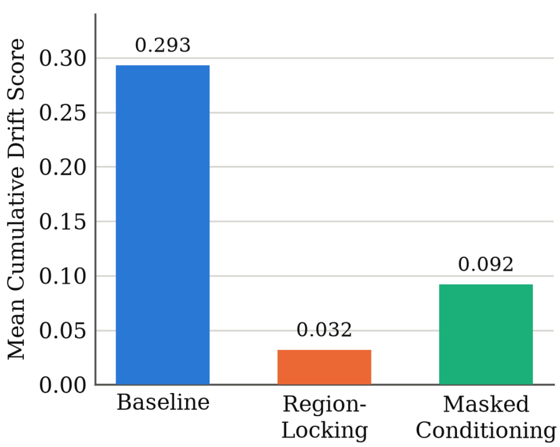

Fig. 3. Mean cumulative Drift Score by condition. Error bars omitted; both reductions are significant at $p < 0.0001$ by paired t-test and Wilcoxon signed-rank test.

Region-locking also outperforms masked conditioning directly, in a head-to-head paired comparison ($p < 0.0001$). Broken down by chain type (Table V), both mitigations perform slightly better on global chains than on object-level chains, though both remain highly significant in each subgroup, suggesting the target-identification weak point discussed in Section IV-C, which is specific to object-adding instructions, affects object-level chains more than global ones. On the worst-case chain from Section III-B, baseline cumulative drift of 0.565 falls to 0.076 under region-locking and 0.180 under masked conditioning, both large, visually confirmed recoveries (Fig. 4).

TABLE V
DRIFT REDUCTION BY CHAIN TYPE ($n = 60$ PER CELL)

Chain Type	Condition	Reduction	Wilcoxon p
Object-Level	Masked Conditioning	62.5%	< 0.0001
Object-Level	Region-Locking	86.0%	< 0.0001
Global	Masked Conditioning	74.8%	< 0.0001
Global	Region-Locking	92.2%	< 0.0001

Fig. 4. The worst-case baseline chain (baseball glove) under all three conditions. The baseline collapses to near-black; region-locking recovers the scene almost exactly, with one visible seam artifact near the glove; masked conditioning recovers the scene for three of four steps but produces an unrelated image on the final "add a cap" instruction (Section IV-A).

D. RQ2: Compounding Across the Chain

No single consecutive step transition is statistically significant (step 1 to 2: $p = 0.85$; 2 to 3: $p = 0.33$; 3 to 4: $p = 0.47$), meaning there is no detectable escalation at any individual point in the chain (Fig. 5). Comparing the two endpoints directly, step 1 versus step 4 is significant ($p = 0.012$ by t-test, $p = 0.006$ by Wilcoxon), but with step 4 showing *less* drift (mean 0.066) than step 1 (mean 0.080), the reverse of the predicted direction.

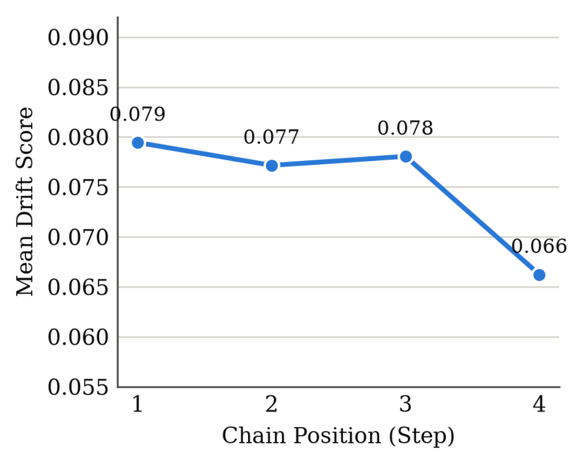

Fig. 5. Mean baseline Drift Score by chain position. The step 1 to step 4 difference is significant ($p = 0.012$); no individual step-to-step transition is.

IV. Discussion

A. Why the Compounding Hypothesis Did Not Hold

The working hypothesis going into this study was that later edits would show progressively more collateral damage, because each one operates on an already-modified image. The data shows the opposite, at the one point where a significant difference exists at all. Table III shows why: in the flagship chain, baseline drift climbs steeply through steps 1 to 3 (0.074, 0.213, 0.278) and then becomes unmeasurable at step 4, not because the model stopped drifting but because the image had already collapsed to near-total black and SAM could no longer find enough regions to score it. A chain that fails this early has, by construction of a CLIP-similarity metric, very little room left to register as further changed in whatever steps remain. An image that is already almost entirely black cannot become much more different from its original self than it already is. This is a measurement ceiling effect masking a front-loaded failure, not evidence that later edits are genuinely gentler, and the aggregate step-position averages in Fig. 5 are consistent with several chains following this same early-collapse pattern rather than drifting gradually. Separating "drift is front-loaded and severe" from "drift does not compound" would need a metric sensitive to further degradation of an already-degraded image, which is a natural direction for follow-up work rather than something the CLIP-similarity approach used here can resolve as built.

B. Why Region-Locking Outperformed Masked Conditioning

The prediction going in was that masked conditioning, a stricter constraint applied at generation time, would outperform region-locking's after-the-fact correction, at some cost to edit quality at the region boundary. The opposite happened on the drift metric itself, and the likely reason is a property of how the metric and the mitigation interact rather than either mitigation being fundamentally weaker. Masked conditioning edits a padded crop, the target box plus a 32-pixel margin, deliberately included so the model has enough surrounding context to produce a coherent edit. The Drift Score, however, only excludes the raw target box from scoring, so masked conditioning's own padding ring is counted as drift even when nothing objectionable happened there. Region-locking's hard revert has no equivalent margin. This should not be read as proof that region-locking produces better-looking edits in general; the qualitative example in Fig. 4 shows region-locking has its own artifact, a visible, unblended rectangular seam near the glove, consistent with the boundary-cost intuition behind the original hypothesis. It simply does not show up as a drift-score cost the way that hypothesis predicted.

C. A Shared Weak Point: Target-Region Identification

Both mitigations, and the Drift Score itself, depend on CLIP correctly identifying which region an instruction targets. This is reliable for "change X" and "remove X" instructions, where the target already exists in the pre-edit image. It is not reliable for "add X" instructions, where nothing pre-existing can match well. Tracing one concrete failure directly: for the instruction "add a cap on the boy's head," the final step of the worst-case chain, the identified target box covered 99.6% of the frame (Fig. 6), essentially the entire picture rather than a small region around the head. Masked conditioning's spatial constraint therefore gave no real protection for that step, since it was cropping to almost the entire image regardless, which is exactly the step 4 spike visible for masked conditioning in Table III. Region-locking faced the identical oversized box on the same step and produced a comparatively smaller spike, most likely because diffusion sampling is stochastic with no fixed seed, so the two mitigations' generation calls for that step were independent random draws from the same model, and one landed more conservative than the other. This is a reproducible limitation of the target-identification approach, not an implementation bug in either mitigation, and it affects roughly half of the object-level chains, those whose instructions use an "add" verb.

Fig. 6. Identified target region (red outline) for "add a cap on the boy's head," covering 99.6% of the frame.

To check whether this single traced case reflects a dataset-wide pattern, every step across all 480 was split into "add" instructions and all others, and compared with an independent-samples Mann-Whitney U test, since add and non-add steps are not paired observations within the same chain (Table VI). Baseline and region-locking both show lower mean drift on add steps than on other steps, a difference that is only borderline significant ($p = 0.052$ and $p = 0.072$ respectively) rather than conclusive. Masked conditioning is the outlier: its add-step mean is numerically *higher* than its other-step mean, the only condition of the three where this reverses, though the difference itself does not reach significance ($p = 0.693$) at this sample size. The direction is consistent with the mechanism traced above, an oversized target box removing masked conditioning's spatial constraint specifically on add instructions, but the dataset-wide test should be read as corroborating, not confirming, the single traced example; a larger sample would be needed to establish the effect conclusively.

TABLE VI
Per-Step Drift by Instruction Type

Condition	"Add" Mean	Other Mean	Mann-Whitney p
Baseline	0.068	0.078	0.052
Region-Locking	0.006	0.009	0.072
Masked Conditioning	0.026	0.022	0.693

D. Limitations

Six constraints bound the present study. The 60-image, 120-chain dataset supports the paired tests reported here but is an initial empirical study, not a definitive benchmark. All results are specific to InstructPix2Pix; model-agnosticism is a design goal, not something tested against a second editor here. CLIP similarity is an imperfect proxy for human-perceived change, and the human-rating tool built for this purpose (15 stratified chains) was not administered, an optional step skipped under the 30-day budget. Target-region identification through CLIP's top-1 similarity is a known weak point for "add" instructions. Region boxes, not exact per-pixel masks, are used throughout, which is the direct cause of masked conditioning's padding-margin effect. Finally, attention-restricted editing was scoped but not implemented, since RQ3 has a clear answer without it.

E. Future Work

The most direct extension is evaluating the Drift Score against a second, more recent editor, to test the model-agnosticism claimed but not exercised here. Running the built human-perceptual check would validate the CLIP-based numbers against human judgment on the same 15 chains. A metric less prone to the ceiling effect in Section IV-A, sensitive to further degradation of an already-degraded image, would let a future study separate gradual compounding from sudden, front-loaded failure with more confidence than the present metric allows.

V. Conclusion

This paper set out to answer whether unintended drift across a chain of instruction-based image edits can be measured automatically, whether it compounds, and whether cheap mitigations reduce it. RQ1 and RQ3 have clear, positive, and statistically well-supported answers: yes, and yes, by a wide margin, with region-locking cutting mean cumulative drift by 89.1% and masked conditioning by 68.6%, both at $p < 0.0001$. RQ2's answer is more interesting than a simple yes or no. The data suggests drift may be front-loaded into occasional catastrophic failures rather than gradually compounding, a pattern the Drift Score as built cannot fully distinguish from no compounding at all. Two specific mechanisms, a measurement ceiling effect and a metric-mitigation margin mismatch, explain why two of the original hypotheses reversed rather than simply failing to replicate, which is itself evidence that the underlying measurement is capturing something real and interpretable rather than noise.

All results are specific to InstructPix2Pix, and the human-perceptual validation step was built but not administered; resolving both, together with testing against a second editor, is the next step toward a benchmark the field can rely on rather than a single-project case study.

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